

Healthcare RAG Bot using Retrieval-Augmented Generation (RAG)

Problem Statement

Healthcare information is vast, complex, and continuously evolving. Patients and healthcare professionals often struggle to quickly access relevant medical information from large collections of documents such as disease descriptions, treatment guidelines, symptom databases, and healthcare manuals. Traditional chatbots rely on pre-trained knowledge and may provide outdated or inaccurate responses when dealing with domain-specific medical queries.

The objective of this project is to develop a Healthcare Retrieval-Augmented Generation (RAG) Bot that can answer healthcare-related questions by retrieving relevant information from a custom medical knowledge base and generating context-aware responses. The system combines document retrieval techniques with large language models to improve response accuracy and reduce hallucinations.

The proposed solution uses FAISS for efficient vector-based document retrieval, Hugging Face Embeddings for converting medical documents into vector representations, LangChain for orchestrating the retrieval pipeline, and Flask for backend API development. When a user submits a query, the system retrieves the most relevant healthcare documents from the knowledge base and uses them as context to generate informative responses.

This project demonstrates the practical application of Retrieval-Augmented Generation in healthcare, enabling users to obtain reliable information from a curated dataset while maintaining fast response times and improved accuracy compared to conventional chatbots.

Objectives

- Build a healthcare-focused question-answering system.
- Implement Retrieval-Augmented Generation (RAG) architecture.
- Retrieve relevant medical information using vector similarity search.
- Generate context-aware responses using retrieved documents.
- Reduce hallucinations commonly observed in standalone language models.
- Provide a user-friendly web interface for healthcare queries.

Technologies Used

Python, Flask, LangChain, FAISS, Hugging Face Embeddings, HTML, CSS, JavaScript

System Workflow

1. Collect healthcare-related documents.
2. Generate embeddings using Hugging Face models.
3. Store embeddings in a FAISS vector database.
4. User submits a healthcare query.
5. Relevant documents are retrieved from FAISS.
6. Retrieved context is passed to the language model.
7. The model generates an accurate response.
8. Response is displayed through the web interface.

Expected Outcome

The Healthcare RAG Bot will provide accurate and context-aware answers to healthcare-related questions by combining information retrieval and language generation techniques.

Conclusion

The Healthcare RAG Bot demonstrates how Retrieval-Augmented Generation can enhance the quality of domain-specific question-answering systems.